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Initial steps for identifying interdisciplinary directions and their potential applications

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Introduction

Universities and funding bodies often publish lists of interdisciplinary topics to demonstrate their support for interdisciplinarity research (IDR) and choice of flavor for interdisciplinary directions. However, recognizing and selecting such topics is challenging due to epistemic discrepancies in understanding interdisciplinarity (Marres & de Rijcke, 2020) and limitations of current scientometric indicators, which often neglect IDR's evolving nature (Zhou et al., 2022). Research can be regarded as interdisciplinarity 20 years earlier yet deemed disciplinary later. For instance, bioinformatics began as an interdisciplinary field in the early 1960s but has since become a distinct discipline. To accurately assess IDR, it's crucial to conceptualize and operationalize interdisciplinarity as a temporal concept, accounting for potential decay in value over time—an element missing in current scientometric indicators of IDR.

This study aims to propose methods of identifying new interdisciplinary directions (topics) based on this temporal understanding of interdisciplinarity. The rest of the paper is organized as follows. We begin by introducing our proposal. We then show some empirical explorations using bibliographic data of papers in Scientometrics from 1991 to 2020. The paper concludes with an extensive discussion of the limitations that the current empirical analyses possess and plans for the next step.

A Prototype for Identifying Interdisciplinary Directions

Our goal is to identify new interdisciplinary directions a) in a certain discipline and b) for the assessed period, therefore allowing time- and discipline-normalized understanding of interdisciplinarity. To be specific, our proposed methods/measurements will be conducted in five steps.

Step 1. Constructing concept-combination temporal networks. First, for a corpus of publications representing a discipline, we construct several concept-combination temporal networks to simulate the evolution of knowledge combinations. The networks take all the combined (co-occurred) concepts (e.g. keywords in titles, abstracts, or keywords in references' titles and abstracts) as nodes and their cooccurrence as edge representing concept combination.

Step 2. Examining the growth of networks. Second, we compare the obtained concept combination pairs between adjacent periods and examine the growth of networks: new nodes or clusters of nodes (concepts), new edges, new communities, etc. One could see such new elements in the concept-combination network as candidates for interdisciplinary directions.

Step 3. Determining the randomness of growth in networks. It is important to determine whether the inclusions of new elements in the concept-combination networks are significant or random, e.g. only appeared once. To assess the "randomness" of inclusions, we will compare the empirical networks with null configuration models and determine if such growth delivers meaningful implications for interdisciplinarity.

Step 4. Identifying interdisciplinary directions. Once meaningful growths in concepts and their combinations are obtained, we can further group them into several interdisciplinary directions through techniques such as network clustering. Each interdisciplinary direction can be operationalized as a list of concepts (keywords) that can be queried to retrieve related publications in the research portfolio of institutions.

Step 5. Example application: evaluating the IDR capacity of research institutions. In the last step, one could evaluate the IDR capacities of research institutions by examining the volume, direction, and synergy of publications related to the identified interdisciplinary directions.

Data and Methodology

To make an initial attempt at the proposed method, we collect bibliographic data for the journal *Scientometrics (SCIM)* from OpenAlex (Priem et al., 2022) for 5,405 publications and their 132,224 references. For each reference, we retrieve the L2 (20k+) concepts assigned to them by OpenAlex as a proxy for aggregated keywords denoting their research topics. In this poster, we explore the first two steps in our proposed prototype: constructing the concept-combination temporal networks and examining the growth in networks.

Preliminary Results: Interdisciplinary Directions in the Journal *Scientometrics* 1991-2020

The distribution of newly cited concepts

In the past 30 years, as SCIM published more papers, increasing new concepts are cited and reached around 500 by 2020. The citation of new concepts exhibits a power-law-like distribution and is more skewed than that of all concepts. 70% of new concepts were cited only once, while "Deep learning" was cited by 45 publications during 2016-2020. Over 40% of SCIM publications cited at least one new concept, with higher proportions before 2010, but the proportion dropped to 20% for publications citing at least three new concepts.

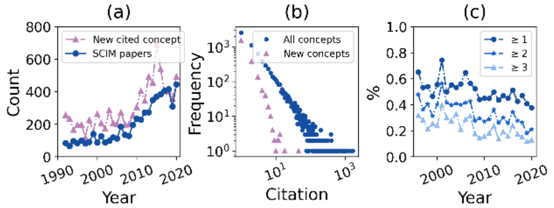


Figure 1. The distribution of newly cited concepts. (a) number of SCIM papers and newly cited concepts; (b) citation distribution for all and newly cited concepts; (c) percentage of papers citing at least 1, 2, or 3 new concepts.

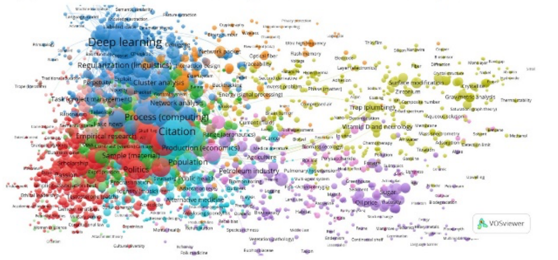
New cited concepts in the knowledge base of Scientometrics

Figure 2c shows examples of highly cited new concepts from 1996-2020, illustrating interdisciplinary evolution in Scientometrics. Early concepts included "the internet" and "corporate governance," followed by network science, nanotechnology, environmental science, and most recently, deep learning and machine learning. We also investigated how new concepts attach to established knowledge bases and potential cluster emergence. Figure 2a&b displays clusters of new concepts and connections to existing concepts. Four major clusters are identified: deep learning & machine learning, general social science, material science & life science, and environmental science & food science. Future research aims to distill more meaningful and precise IDR directions in Steps 3 & 4.

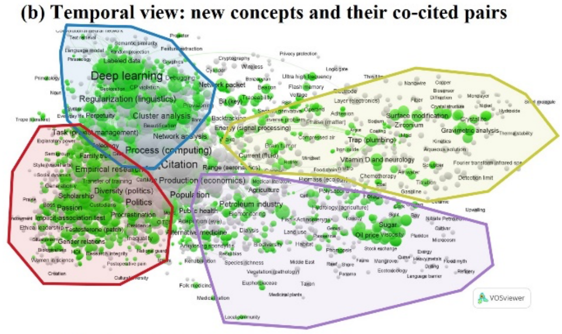
Limitations and Next Steps

This paper has many limitations that we hope to address in future research. Firstly, all concepts for publications are treated equally without considering the association likelihood and their originated discipline. Secondly, the growth of networks is not filtered to obtain meaningful clusters, as suggested in Step 3. Many "randomly cited" concepts remain in the networks. Thirdly, only publications in the journal Scientometrics are studied, which could be insufficient to represent a discipline. In future research, we hope to complete all the analyses proposed in the prototype and provide an in-depth exploration of the potential applications.

(a) Cluster view: new concepts and their co-cited pairs



(b) Temporal view: new concepts and their co-cited pairs



(c) Newly cited concepts

1996 - 2000
The Internet (15)
Corporate governance (13)
Reflexivity (11)
Gatekeeping (9)
Monopoly (8)

2001 - 2005
Dynamic network analysis (11)
Civilization (9)
Knowledge graph (9)
Download (8)
Network formation (8)

2006 - 2010
Attraction (25)
Residence (15)
Societal impact of nanotech (15)
Geodesic (11)
Open research (11)

2011 - 2015
Translational science (18)
Municipal solid waste: (16)
Exergy (16)
Water resources (15)
Wetland (15)

2016 - 2020
Deep learning (45)
Regularization (linguistics) (18)
Ecstasy (13)
Encoder (13)
Green innovation (11)

Figure 2. New cited concepts in the knowledge base of Scientometrics. (a) examples of highly cited new concepts ; (b) Clusters in newly cited concepts and their pairs; (c) The coupling of new concepts (green) and their pairs (grey).

Acknowledgments

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